Digital Electronics - Lecture Notes

1. Boolean Algebra and Logic Gates

The basic gates are AND, OR and NOT. The universal gates are NAND and NOR, each of which can implement any Boolean function. XOR (exclusive-OR) outputs 1 only when its two inputs differ; XNOR is its complement. De Morgan's theorems are: NOT(A AND B) = (NOT A) OR (NOT B) and NOT(A OR B) = (NOT A) AND (NOT B).

A minterm is a product term containing every variable once. Sum-of-products (SOP) form is a sum of minterms; product-of-sums (POS) is a product of maxterms. A Karnaugh map is a rectangular grid used to simplify Boolean expressions; adjacent cells that differ in exactly one variable can be grouped in powers of two (1, 2, 4, 8, ...).

2. Combinational Logic

A half adder adds two bits and produces Sum = A XOR B and Carry = A AND B. A full adder adds three bits (A, B and carry-in) and produces Sum and carry-out; it can be built from two half adders and an OR gate. A multiplexer (MUX) with 2^n data inputs needs n select lines. A decoder with n inputs has 2^n outputs.

3. Sequential Logic and Flip-Flops

Sequential circuits have memory. A latch is level-sensitive; a flip-flop is edge-triggered. The RS latch has an invalid state when both inputs are asserted. The D flip-flop stores the input on a clock edge: Q(next) = D. The JK flip-flop removes the invalid state: when J = K = 1 the output toggles. The T flip-flop toggles when T = 1 and holds when T = 0.

Setup time is the interval before the active clock edge during which the data input must be stable. Hold time is the interval after the edge during which the input must remain stable. Violating these causes metastability.

4. Counters and Registers

A mod-N counter has N distinct states and divides the input frequency by N. An n-bit binary ripple counter divides by 2^n. A ring counter circulates a single 1 through a shift register; a Johnson (twisted-ring) counter has 2n states for n flip-flops. A shift register moves data one position per clock and can convert between serial and parallel formats.